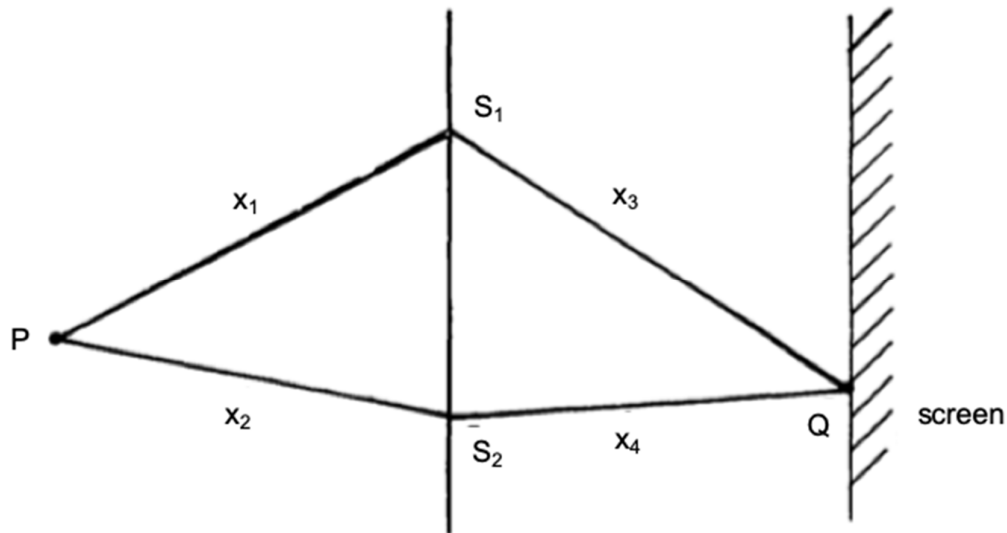


magnitude, which is  $x_0$ . Therefore, Option D is wrong.

- 19 Two identical narrow slits  $S_1$  and  $S_2$  are illuminated with light of wavelength  $\lambda$  from a point source P.



If, as shown in the diagram above, the light is then allowed to fall on a screen, and if  $m$  is a positive integer, the condition for constructive interference at Q is

- |   |                                        |
|---|----------------------------------------|
| A | $x_1 - x_2 = (2m+1) \frac{\lambda}{2}$ |
| B | $x_3 - x_4 = (2m+1) \frac{\lambda}{2}$ |
| C | $x_3 - x_4 = m\lambda$                 |
| D | $(x_1 + x_3) - (x_2 + x_4) = m\lambda$ |

**Answer: D**

|  |                                                                                                                                                                                                                                                                      |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | For constructive interference to take place, the waves meeting at the point must have a path difference (difference in the lengths of paths from the source to the meeting point) to be a full wavelength, or integers of multiples of the wavelengths of the waves. |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|